

Basic Electrical Engineering (BEE) - Unit 1 Concepts

What is Ohm's Law?

Ohm's Law states that the voltage (V) across a resistor is directly proportional to the current (I) flowing through it. The constant of proportionality is the resistance (R).

The formula is $V = I * R$.

What is Kirchhoff's Current Law (KCL)?

KCL states that the total current entering a junction or node is exactly equal to the total current leaving the node. This is based on the conservation of charge.

What is Kirchhoff's Voltage Law (KVL)?

KVL states that in any closed loop or mesh in a circuit, the algebraic sum of all the voltages (rises and drops) is equal to zero. This is based on the conservation of energy.

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Basic Electrical Engineering (BEE) - Unit 2 Concepts

What are Resistors in Series?

When resistors are connected end-to-end (in series), the total resistance (R_{total}) is the sum of their individual resistances.

The formula is $R_{total} = R_1 + R_2 + R_3 + \dots$

The current is the same through all resistors in series.

What are Resistors in Parallel?

When resistors are connected across the same two points (in parallel), the reciprocal of the total resistance is the sum of the reciprocals of the individual resistances.

The formula is $1 / R_{total} = 1 / R_1 + 1 / R_2 + 1 / R_3 + \dots$

The voltage is the same across all resistors in parallel.

What is a Transformer?

A transformer is a passive electrical device that transfers electrical energy from one circuit to another through electromagnetic induction. It is most commonly used to increase ('step-up') or decrease ('step-down') voltage levels.